

Different types of equipment for tillage...

CAE1.3 — Introduction to Agricultural Engineering

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Different types of equipment for tillage, sowing, planting and transplanting, 4 fertilizer application, weed control, and plant protection; Harvesting and threshing equipment for rice, wheat, maize, cotton, sugarcane, fruits, tuber crops and other locally important crops

Definition

Different types of equipment are used for tillage, sowing, planting, transplanting, fertilizer application, weed control, plant protection, and harvesting and threshing of various crops like rice, wheat, maize, cotton, sugarcane, fruits, tuber crops, and other locally important crops. These machines are essential for efficient and mechanized agricultural practices.

KEY POINT Agricultural equipment is tailored to the specific needs of each crop and operation, ensuring optimal productivity and resource use.

Key concepts

- **Tillage equipment:** Tools like plows, cultivators, and harrows are used to prepare the soil for planting. Primary tillage (e.g., plowing) breaks up the soil, while secondary tillage (e.g., harrowing) smooths the surface.
- **Sowing equipment:** Seed drills and planters distribute seeds evenly in rows, ensuring uniform spacing and depth for optimal germination.
- **Planting equipment:** Transplanters and seeders are used to plant seedlings or seeds in the field, often with precision to ensure proper crop growth.
- **Fertilizer application equipment:** Spreaders and applicators distribute fertilizers evenly across the field, ensuring nutrient availability for crops.
- **Weed control equipment:** Cultivators and hoes are used to remove weeds by uprooting or burying them, reducing competition with crops.
- **Plant protection equipment:** Sprayers and dusters apply pesticides and herbicides to protect crops from pests and diseases.
- **Harvesting equipment:** Machines like combines, reapers, and threshers are used to cut, gather, and separate crops from their stalks.
- **Threshing equipment:** Threshers and separators remove grains from the stalks, often used in conjunction with harvesters for crops like rice and wheat.

Comparison

Basis	Tillage Equipment	Sowing Equipment
Function	Prepares soil for planting	Distributes seeds evenly
Examples	Plows, cultivators, harrows	Seed drills, planters
Depth	Can vary from shallow to deep	Typically shallow

Basis	Fertilizer Application Equipment	Weed Control Equipment
Function	Applies fertilizers to crops	Removes weeds manually or mechanically
Examples	Spreaders, applicators	Cultivators, hoes
Method	Spraying or spreading	Uprooting or burying

Basis	Harvesting Equipment	Threshing Equipment
Function	Cuts and gathers crops	Separates grains from stalks
Examples	Combines, reapers	Threshers, separators
Crop Specific	Rice, wheat, maize, sugarcane	Rice, wheat, barley

How it works

Tillage equipment, such as plows and cultivators, break up compacted soil to improve aeration and water infiltration. Primary tillage tools like plows create a rough surface, while secondary tools like harrows smooth it for seedbed preparation. Sowing equipment, such as seed drills, ensures seeds are placed at the correct depth and spacing for uniform germination. Fertilizer application equipment, like spreaders, ensures nutrients are evenly distributed across the field. Weed control equipment, such as cultivators, disrupts weed growth by uprooting or burying them, reducing competition with crops. Plant protection equipment, like sprayers, applies chemicals to control pests and diseases. Harvesting equipment, such as combines, cuts and gathers crops in one operation, while threshing equipment separates grains from the stalks, often used for crops like rice and wheat.

INDIAN CONTEXT In Indian agriculture, small and marginal farmers often use manual or low-cost equipment, while large farms use mechanized systems. For example, rice is harvested using combine harvesters in states like Punjab and West Bengal, while sugarcane is cut manually in states like Uttar Pradesh and Maharashtra.



Barn and farm equipment at Dagger Gate - [geograph.org.uk](https://www.geograph.org.uk) - 763124 — Nigel Chadwick, CC BY-SA 2.0



Strip tillage disk-row equipment is vital to healthy cropland in Knox County, Texas. (24743091209) — USDA NRCS Texas, Public domain

Exam pointers

- List the types of equipment used for each agricultural operation (tillage, sowing, etc.).
- Explain the difference between primary and secondary tillage.
- Describe the function of a planter and a sprayer.
- Mention the types of machines used for harvesting and threshing rice and wheat.

COMMON MISTAKE Students often confuse the functions of tillage and sowing equipment. Remember: tillage prepares the soil, while sowing places the seeds.

Check yourself

Question: What is the main function of secondary tillage?

Answer: Secondary tillage smooths the soil surface and prepares a fine seedbed for planting.

Question: Name two types of equipment used for sowing.

Answer: Seed drills and planters.

Question: What is the purpose of a sprayer in agriculture?

Answer: A sprayer is used to apply pesticides, herbicides, or fertilizers to crops.

Question: Which equipment is used to separate grains from the stalks?

Answer: Threshing equipment such as threshers or separators.

Sources consulted

1. [Tillage](#) — Wikipedia, CC BY-SA 4.0
2. [Agricultural machinery](#) — Wikipedia, CC BY-SA 4.0
3. [Glossary of agriculture](#) — Wikipedia, CC BY-SA 4.0
4. [Horticulture/Printable version](#) — Wikibooks, CC BY-SA 4.0
5. Image: [Barn and farm equipment at Dagger Gate - geograph.org.uk - 763124](#) — Nigel Chadwick, CC BY-SA 2.0, via Wikimedia Commons
6. Image: [Strip tillage disk-row equipment is vital to healthy cropland in Knox County, Texas. \(24743091209\).jpg](#) — USDA NRCS Texas, Public domain, via Wikimedia Commons

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